

CHAPTER 4

Animal Kingdom

Basis of Classification

- Which of the following pairs is an **incorrect** match? (2024 Re)
 - Annelids and arthropods - Bilateral symmetry
 - Sponges - Acoelomates
 - Coelenterates and Ctenophores - Radial symmetry
 - Platyhelminthes - Diploblastic organisation
- Consider the following statements: (2024)
 - Annelids are true coelomates
 - Poriferans are pseudocoelomates
 - Aschelminthes are acoelomates
 - Platyhelminthes are pseudocoelomates
 Choose the correct answer from the options given below :
 - C only
 - D only
 - B only
 - A only
- Bilaterally symmetrical and acoelomate animals are exemplified by: (2020)
 - Platyhelminthes
 - Aschelminthes
 - Annelida
 - Ctenophora
- Consider following features
 - Organ system level of organisation
 - Bilateral symmetry
 - True coelomates with segmentation of body
 Select the correct option of animal groups which possess all the above characteristics (2019)
 - Annelida, Arthropoda and Chordata
 - Annelida, Arthropoda and Mollusca
 - Arthropoda, Mollusca and Chordata
 - Annelida, Mollusca and Chordata

Porifera, Coelenterata, Ctenophora, Platyhelminthes, Aschelminthes

- Select the **correct** statements: (2023-Manipur)
 - Platyhelminthes are triploblastic pseudocoelomate and bilaterally symmetrical organisms.
 - Ctenophores reproduce only sexually and fertilization is external.
 - In tapeworm, fertilization is internal but sexes are not separate.

D. Ctenophores are exclusively marine, diploblastic and bioluminescent organisms.

E. In sponges, fertilization is external and development is direct.

Choose the **correct** answer from the options given below:

- (A), (C) and (D) only
 - (B), (C) and (D) only
 - (A) and (E) only
 - (B) and (D) only
- In case of poriferans, the spongocoel is lined with flagellated cells called: (2017-Delhi)
 - Ostia
 - Oscula
 - Choanocytes
 - Mesenchymal cells
 - Which of the following endoparasites of humans does show viviparity? (2015)
 - Trichinella spiralis*
 - Ascaris lumbricoides*
 - Ancylostoma duodenale*
 - Enterobius vermicularis*
 - Metagenesis refers to: (2015 Re)
 - Alternation of generation between asexual and sexual phases of organisms
 - Occurrence of a drastic change in form during post-embryonic development
 - Presence of a segmented body and parthenogenetic mode of reproduction
 - Presence of different morphic forms
 - Planaria* possess high capacity of: (2014)
 - Bioluminescence
 - Metamorphosis
 - Regeneration
 - Alternation of generation

Annelida, Arthropoda, Mollusca, Echinodermata

- Match List-I with List-II. (2023)

List-I		List-II	
(A)	<i>Taenia</i>	(I)	Nephridia
(B)	<i>Paramoecium</i>	(II)	Contractile vacuole
(C)	<i>Periplaneta</i>	(III)	Flame cells
(D)	<i>Pheretima</i>	(IV)	Urecose gland

Choose the correct answer from the options give below:

- A-II, B-I, C-IV, D-III
- A-I, B-II, C-III, D-IV
- A-I, B-II, C-IV, D-III
- A-III, B-II, C-IV, D-I

11. Match List-I with List-II. (2023-Manipur)

List-I		List-II	
(A)	Contractile vacuole	(I)	<i>Asterias</i>
(B)	Water vascular system	(II)	<i>Amoeba</i>
(C)	Canal system	(III)	<i>Spongilla</i>
(D)	Flame cells	(IV)	<i>Taenia</i>

Choose the **correct** answer from the options given below:

- a. A-IV, B-II, C-I, D-III b. A-I, B-III, C-II, D-IV
c. A-III, B-II, C-I, D-IV d. A-II, B-I, C-III, D-IV

12. Match List-I with List-II (2021)

List-I		List-II	
(A)	Metamerism	(i)	Coelenterata
(B)	Canal system	(ii)	Ctenophora
(C)	Comb plates	(iii)	Annelida
(D)	Cnidoblasts	(iv)	Porifera

Choose the correct answer from the options given below.

- a. A-iii B-iv C-i D-ii
b. A-iii B-iv C-ii D-i
c. A-iv B-i C-ii D-iii
d. A-iv B-iii C-i D-ii

13. Match the following: (2021)

List-I		List-II	
(A)	<i>Physalia</i>	(i)	Pearl oyster
(B)	<i>Limulus</i>	(ii)	Portuguese Man of War
(C)	<i>Ancylostoma</i>	(iii)	Living fossil
(D)	<i>Pinctada</i>	(iv)	Hookworm

Choose the correct answer from the options given below.

- a. A-iv B-i C-iii D-ii
b. A-ii B-iii C-iv D-i
c. A-i B-iv C-iii D-ii
d. A-ii B-iii C-i D-iv

14. Read the following statements. (2021)

- A. Metagenesis is observed in Helminths.
B. Echinoderms are triploblastic and coelomate animals.
C. Round worms have organ-system level of body organization.
D. Comb plates present in ctenophores help in digestion.
E. Water vascular system is characteristic of Echinoderms.

Choose the correct answer from the options given below.

- a. A, B and C are correct b. A, D and E are correct
c. B, C and E are correct d. C, D and E are correct

15. Match the following columns and select the correct option. (2020)

Column-I		Column-II	
(A)	Gregarious, polyphagous pest	(i)	<i>Asterias</i>
(B)	Adult with radial symmetry and larva with bilateral symmetry	(ii)	Scorpion
(C)	Book lungs	(iii)	<i>Ctenoplana</i>
(D)	Bioluminescence	(iv)	<i>Locusta</i>

- a. A-iv B-i C-ii D-iii
b. A-iii B-ii C-i D-iv
c. A-ii B-i C-iii D-iv
d. A-i B-iii C-ii D-iv

16. Which of the following options does correctly represent the characteristic features of phylum Annelida? (2020-Covid)

- a. Triploblastic, segmented body and bilaterally symmetrical
b. Triploblastic, flattened body and acoelomate condition
c. Diploblastic, mostly marine and radially symmetrical
d. Triploblastic, unsegmented body and bilaterally symmetrical

17. Match the following organisms with their respective characteristics : (2019)

- A. *Pila* i. Flame cells
B. *Bombyx* ii. Comb plates
C. *Pleurobrachia* iii. Radula
D. *Taenia* iv. Malpighian tubules

Select the correct option from the following:

- a. A-iii B-ii C-i D-iv
b. A-iii B-iv C-ii D-i
c. A-ii B-iv C-iii D-i
d. A-iii B-ii C-iv D-i

18. Which of the following animals does not undergo metamorphosis? (2018)

- a. Earthworm b. Tunicate
c. Moth d. Starfish

19. Which of the following features is not present in the Phylum-Arthropoda? (2016-I)

- a. Chitinous exoskeleton b. Metameric segmentation
c. Parapodia d. Jointed appendages

20. Which of the following characteristics is mainly responsible for diversification of insects on land? (2015)

- a. Exoskeleton b. Eyes
c. Segmentation d. Bilateral symmetry

21. Body having mesh-work of cell, internal cavities lined with food filtering flagellated cells and indirect development are the characteristics of phylum: (2015 Re)

- a. Porifera b. Mollusca
c. Protozoa d. Coelenterate

22. One of the representatives of Phylum Arthropoda is: (2013)

- a. Flying fish b. Cuttlefish
c. Silverfish d. Pufferfish

23. Which of the following are correctly matched with respect to their taxonomic classification? (2013)

- a. Spiny anteater, sea urchin, sea cucumber – Echinodermata
b. Flying fish, cuttlefish, silverfish, – Pisces
c. Centipede, millipede, spider, scorpion – Insecta
d. House fly, butterfly, tsetsefly, silverfish – Insecta

24. Which group of animals belongs to the same phylum? (2013)
- Sponge, Sea anemone, Starfish
 - Malarial parasite, *Amoeba*, Mosquito
 - Earthworm, Pinworm, Tapeworm
 - Prawn, Scorpion, *Locusta*

Hemichordata and Chordata

25. Given below are two statements: One is labelled as **Assertion A** and the other is labelled as **Reason R**. (2024 Re)

Assertion A: Members of subphylum vertebrata possess notochord during the embryonic period. The notochord is replaced by a cartilaginous or bony vertebral column in the adult.

Reason R: Thus all chordates are vertebrates but not all vertebrates are chordates.

In the light of the above statements, choose the **correct** answer from the options given below.

- A is true but R is false.
 - A is false but R is true.
 - Both A and R are true and R is the correct explanation of A.
 - Both A and R are true but R is NOT the correct explanation of A.
26. Open Circulatory system is present in: (2024 Re)
- Palaemon*, *Nereis*, *Balanoglossus*
 - Hirudinaria*, *Bombyx*, *Salpa*
 - Anopheles*, *Limax*, *Limulus*
 - Pheretima*, *Musca*, *Pila*
27. The following are the statements about non-chordates: (2024)
- Pharynx is perforated by gill slits.
 - Notochord is absent.
 - Central nervous system is dorsal.
 - Heart is dorsal if present.
 - Post anal tail is absent.
- Choose the most appropriate answer from the options given below:
- B, D & E only
 - B, C & D only
 - A & C only
 - A, B & D only

28. Radial symmetry is NOT found in adults of phylum (2023)
- Echinodermata
 - Ctenophora
 - Hemichordata
 - Coelenterata

29. Select the correct statements with reference to chordates. (2023)

- Presence of a mid-dorsal, solid and double nerve cord.
- Presence of closed circulatory system.
- Presence of paired pharyngeal gills
- Presence of dorsal heart
- Triploblastic pseudocoelomate animals.

Choose the correct answer from the options given below:

- C, D and E only
- A, C and D only
- B and C only
- B, D and E only

30. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R). (2022)

Assertion (A): All vertebrates are chordates but all chordates are not vertebrates.

Reason (R): Notochord is replaced by vertebral column in the adult vertebrates

In the light of the above statements, choose the correct answer from the options given below.

- (A) is not correct but (R) is correct
 - Both (A) and (R) are correct and (R) is the correct explanation of (A)
 - Both (A) and (R) are correct but (R) is not the correct explanation of (A)
 - (A) is correct but (R) is not correct
31. Which of the following statements are true for the phylum-Chordata? (2020)
- In Urochordata notochord extends from head to tail and it is present throughout their life.
 - In Vertebrata notochord is present during the embryonic period only.
 - Central nervous system is dorsal and hollow.
 - Chordata is divided into 3 subphyla: Hemichordata, Tunicata and Cephalochordata.

32. All vertebrates are chordates but all chordates are **not** vertebrates, why? (2020-Covid)

- Ventral hollow nerve cord remains throughout life in some chordates
- all chordates possess vertebral column
- All chordates possess notochord throughout their life
- Notochord is replaced by vertebral column in adult of some chordates

33. An important characteristic that hemichordates share with chordates is: (2017-Delhi)

- Absence of notochord
- Ventral tubular nerve cord
- Pharynx with gill slits
- Pharynx without gill slits

34. Select the taxon mentioned that represents both marine and fresh water species: (2014)

- Cnidaria*
- Echinoderms*
- Ctenophora
- Cephalochordata*

Class Cyclostomata, Chondrichthyes, Osteichthyes

35. Match List-I with List-II: (2024)

List-I		List-II	
(A)	<i>Pleurobrachia</i>	(I)	Mollusca
(B)	Radula	(II)	Ctenophora
(C)	Stomochord	(III)	Osteichthyes
(D)	Air bladder	(IV)	Hemichordata

Choose the correct answer from the options given below:

- A-(II), B-(IV), C-(I), D-(III)
- A-(IV), B-(III), C-(II), D-(I)
- A-(IV), B-(II), C-(III), D-(I)
- A-(II), B-(I), C-(IV), D-(III)

36. Match List-I with List-II:

(2024)

List-I		List-II	
(A)	<i>Pterophyllum</i>	(I)	Hag fish
(B)	<i>Myxine</i>	(II)	Saw fish
(C)	<i>Pristis</i>	(III)	Angel fish
(D)	<i>Exocoetus</i>	(IV)	Flying fish

Choose the correct answer from the options given below:

- a. A-(IV), B-(I), C-(II), D-(III)
 b. A-(III), B-(II), C-(I), D-(IV)
 c. A-(II), B-(I), C-(III), D-(IV)
 d. A-(III), B-(I), C-(II), D-(IV)

37. Select the incorrect statements with respect to Cyclostomes.

(2022 Re)

- A. They lack scales and paired fins.
 B. They have circular mouth with jaws.
 C. They bear 6-15 pairs of gills.
 D. They migrate to deep sea for spawning.

Choose the most appropriate answer from the options given below:

- a. (A) and (D) only b. (A) and (B) only
 c. (B) and (C) only d. (B) and (D) only

38. Match the following columns and select the correct option.

(2020)

Column-I		Column-II	
(A)	6-15 pairs of gill slits	(i)	<i>Trygon</i>
(B)	Heterocercal caudal fin	(ii)	<i>Cyclostomes</i>
(C)	Air bladder	(iii)	<i>Chondrichthyes</i>
(D)	Poison sting	(iv)	<i>Osteichthyes</i>

- A B C D A B C D
 a. (iii) (iv) (i) (ii) b. (i) (iv) (iii) (ii)
 c. (iv) (ii) (iii) (i) d. (ii) (iii) (iv) (i)

39. A jawless fish, which lays eggs in freshwater and whose ammocoetes larvae after metamorphosis return to the ocean is:

(2015 Re)

- a. *Myxine* b. *Neomyxine*
 c. *Petromyzon* d. *Eptatretus*

40. A marine cartilaginous fish that can produce electric current is:

(2014)

- a. *Scoliodon* b. *Pristis* c. *Torpedo* d. *Trygon*

Class Amphibia, Reptilia, Aves, Mammalia

41. Match List-I with List-II.

(2024 Re)

List-I		List-II	
(A)	Chondrichthyes	(I)	<i>Clarias</i>
(B)	Cyclostomata	(II)	<i>Carcharodon</i>
(C)	Osteichthyes	(III)	<i>Myxine</i>
(D)	Amphibia	(IV)	<i>Ichthyophis</i>

Choose the correct answer from the options given below:

- a. A-II, B-IV, C-I, D-III b. A-I, B-III, C-II, D-IV
 c. A-II, B-III, C-I, D-IV d. A-I, B-II, C-III, D-IV

42. The unique mammalian characteristics are: (2023)

- a. pinna, monocondylic skull and mammary glands
 b. hairs, tympanic membrane and mammary glands
 c. hairs, pinna and mammary glands
 d. hairs, pinna and indirect development

43. Which of the following animals has three chambered heart?

(2022 Re)

- a. *Pteropus* b. *Scoliodon*
 c. *Hippocampus* d. *Chelone*

44. In which of the following animals, digestive tract has additional chambers like crop and gizzard? (2022)

- a. *Pavo*, *Psittacula*, *Corvus*
 b. *Corvus*, *Columba*, *Chameleon*
 c. *Bufo*, *Balaenoptera*, *Bangarus*
 d. *Catla*, *Columba*, *Crocodilus*

45. Which one of the following organisms bears hollow and pneumatic long bones? (2021)

- a. *Hemidactylus* b. *Macropus*
 c. *Ornithorhynchus* d. *Neophron*

46. Match the following columns and select the correct option:

(2020-Covid)

Column-I		Column-II	
(A)	<i>Aptenodytes</i>	(i)	Flying fox
(B)	<i>Pteropus</i>	(ii)	Angel fish
(C)	<i>Pterophyllum</i>	(iii)	Lamprey
(D)	<i>Petromyzon</i>	(iv)	Penguin

- A B C D A B C D
 a. (iii) (iv) (i) (ii) b. (iv) (i) (ii) (iii)
 c. (ii) (i) (iv) (iii) d. (iii) (iv) (ii) (i)

47. Match the following group of organisms with their respective distinctive characteristics and select the correct option:

(2020-Covid)

Organisms		Characteristics	
(A)	Platyhelminthes	(i)	Cylindrical body with no segmentation
(B)	Echinoderms	(ii)	Warm blooded animals with direct development
(C)	Hemichordates	(iii)	Bilateral symmetry with incomplete digestive system
(D)	Aves	(iv)	Radial symmetry with indirect development

- A B C D A B C D
 a. (ii) (iii) (iv) (i) b. (i) (ii) (iii) (iv)
 c. (iv) (i) (ii) (iii) d. (iii) (iv) (i) (ii)

48. Which one of these animals is not a homeotherm? (2018)

- a. *Macropus* b. *Chelone* c. *Camelus* d. *Psittacula*

49. Identify the vertebrate group of animals characterised by crop and gizzard in its digestive system. (2018)

- a. Amphibia b. Reptilia
c. Aves d. Osteichthyes

50. Which among these is the correct combination of aquatic mammals? (2017-Delhi)

- a. Seals, Dolphins, Sharks b. Dolphins, Seals, Trygon
c. Whales, Dolphins, Seals d. Trygon, Whales, Seals

51. Which of the following represents order of 'Horse'? (2017-Delhi)

- a. Equidae b. Perissodactyla
c. Caballus d. Ferus

52. Choose the correct statement: (2016-II)

- a. All reptiles have a three-chambered heart.
b. All pisces have gills covered by an operculum.
c. All mammals are viviparous.
d. All cyclostomes do not possess jaws and paired fins.

53. Which one of the following characteristic is not shared by birds and mammals? (2016-I)

- a. Ossified endoskeleton b. Breathing using lungs
c. Viviparity d. Warm blooded nature

54. Which of the following characteristic features always holds true for the corresponding group of animals? (2016-I)

a.	Cartilaginous endoskeleton	Chondrichthyes
b.	Viviparous	Mammalia
c.	Possess a mouth with an upper and a lower jaw	Chordata
d.	3-chambered heart with one incompletely divided ventricle	Reptilia

55. Which is the National Aquatic Animal of India? (2016-I)

- a. Gangetic shark b. River dolphin
c. Blue whale d. Sea-horse

56. Which of the following animals is not viviparous? (2015)

- a. Platypus b. Whale
c. Flying fox (bat) d. Elephant

57. Which of the following represents the correct combination without any exception? (2015)

Characteristics		Class
a.	Sucking and circular mouth; jaws absent, integument without scales; paired appendages	Cyclostomata
b.	Body covered with feathers; skin moist and glandular; fore-limbs form wings; lungs with air sacs	Aves
c.	Mammary glands; hair on body; pinnae; two pairs of limbs	Mammalia
d.	Mouth ventral; gills without operculum; skin with placoid scales, persistent notochord	Chondrichthyes

58. Match the name of the animal (Column-I) with one characteristics (Column-II) and the phylum/class (Column -III) to which it belongs: (2013)

Column-I	Column-II	Column-III
a. <i>Adamsia</i>	Radially symmetrical	Porifera
b. <i>Petromyzon</i>	Ectoparasite	Cyclostomata
c. <i>Ichthyophis</i>	Terrestrial	Reptilia
d. <i>Limulus</i>	Body covered by chitinous exoskeleton	Pisces

Answer Key

1. (d) 2. (d) 3. (a) 4. (a) 5. (b) 6. (c) 7. (a) 8. (a) 9. (c) 10. (d)
11. (d) 12. (b) 13. (b) 14. (c) 15. (a) 16. (a) 17. (b) 18. (a) 19. (c) 20. (a)
21. (a) 22. (c) 23. (d) 24. (d) 25. (a) 26. (c) 27. (a) 28. (c) 29. (c) 30. (b)
31. (c) 32. (d) 33. (c) 34. (a) 35. (d) 36. (d) 37. (d) 38. (d) 39. (c) 40. (c)
41. (c) 42. (c) 43. (d) 44. (a) 45. (d) 46. (b) 47. (d) 48. (b) 49. (c) 50. (c)
51. (b) 52. (d) 53. (c) 54. (a) 55. (b) 56. (a) 57. (a) 58. (b)